

FreeFusion: Infrared and Visible Image Fusion via Cross Reconstruction Learning

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Abstract—Existing fusion methods empirically design elaborate fusion losses to retain the specific features from source images. Since image fusion has no ground truth, the hand-crafted losses may not make the fused images cover all the vital features, and then affect the performance of the high-level tasks. Here, there are two main challenges: domain discrepancy among source images and semantic mismatch at different-level tasks. This paper proposes an infrared and visible image fusion via cross reconstruction learning, which doesn't use any hand-crafted fusion losses, but prompts the network to adaptively fuse complementary information of source images. Firstly, we design a cross reconstruction learning model that decouples the fusion features to reconstruct another-modality source image. Thus, the fusion network is forced to learn the domain-adaptive representations of two modal features, which enables their domain alignment in a latent space. Secondly, we propose a dynamic interactive fusion strategy that builds a correlation matrix between fusion features and object semantic features to overcome the semantic mismatch. Further, we enhance the strong correlation features and suppress the weak correlation features to improve the interactive ability. Extensive experiments on three datasets demonstrate the superior fusion performance compared to the state-of-the-art methods, concurrently facilitating the segmentation accuracy.

Index Terms—Infrared and visible image fusion, cross reconstruction learning, dynamic interaction fusion, feature correlation matrix.

I. INTRODUCTION

VISIBLE images are obtained by reflected light information, containing rich edge and texture details. But they are sensitive to light conditions and adverse weather. Infrared images are captured through thermal radiation information, effectively highlighting objects in nighttime and smoky conditions. But they lack sufficient details. Infrared and visible image fusion (IVIF) aims to integrate their complementary information that generates a new image with strong perceptual capability,

thereby benefiting the subsequent high-level vision tasks, e.g., semantic segmentation [4], [5], object detection [6], and object tracking [7].

Existing IVIF methods [8], [9], [10], [11], [12] can be divided into four categories. (1) Encoder-decoder based methods (e.g., YDTR [13], U2Fusion [14], PIAFusion [15], CDDFuse [16]) design modality specific encoders to extract information from the source images. Then, the fusion feature representation is generated by fusion modules (e.g., multi-scale feature fusion layer or attention mechanism). Finally, the decoders are used to reconstruct the fusion image. (2) Adversarial learning based methods (e.g., FusionGAN [17], DDcGAN [18], TGFuse [19]) design generators to map condition vectors to the fusion image space, and discriminators constrain the fusion image to retain thermal radiation information and texture information. (3) Joint multi-task based methods (e.g., Tardal [1], MetaFusion [20], SegMif [21]) utilize high-level tasks to provide object semantic features, thereby prompting the fusion features to obtain object contour information. (4) Autoencoder-based methods (e.g., DIDFuse [22], NestFuse [23]) train an autoencoder that obtains the ability of extracting image features by reconstructing corresponding to image. Then, hand-crafted fusion rules (e.g., l_1 -norm, element-wise addition and element-wise choose-max) are designed to fuse complementary features. For example, NestFuse [23] adopts spatial/channel attentional fusion strategy to manually fuse features. Although autoencoder-based methods are not designed with explicit fusion losses, the hand-crafted fusion rules cannot adaptively fuse the important information of two modalities. In contrast, we decouple the fusion feature into two modality features, and then inversely map them to the source images that forces the fusion features to adaptively contain all the source image information. Moreover, since IVIF has no ground truth, the previous three categories of methods construct various hand-crafted fusion loss functions for feature extraction, feature fusion and image reconstruction. As shown in Fig. 1(a), pixel intensity losses (e.g., L_1 [1], [15], L_2 [13], [14], [16], [21], etc.) constrain infrared and visible encoders to extract thermal radiation and texture features. Structural losses (e.g., $SSIM$ [1], [13], [14], [16], [21], SF [13], etc.) ensure that the feature fusion module contains the structural and edge information. They both help the fusion image decoder generate the fusion image. More loss functions are shown in Fig. 2(Left). However, constructing elaborate loss functions may not effectively supervise IVIF framework to retain all the helpful information of source images.

Especially, there are two challenges to IVIF. Domain discrepancy between infrared and visible features and semantic

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Our code is publicly available at <https://github.com/HengshuaiCui/FreeFusion>.

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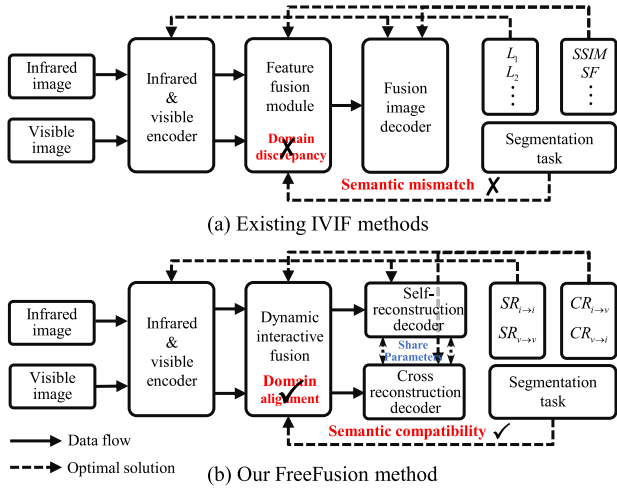


Fig. 1. Motivation of the proposed FreeFusion. Existing IVIF methods design pixel intensity and structural losses that retain the helpful information of source images. But they fail to adequately address the domain discrepancy between source images. Moreover, high-level task is integrated into the fusion task through embedding strategies (e.g., hierarchical interaction [1], bi-level formulation [2], progressive cycle attention [3], etc.). But the two different-level tasks have semantic mismatch. In contrast, without designing any fusion losses, we utilize $SR_{i \rightarrow i}$, $SR_{v \rightarrow v}$, $CR_{i \rightarrow v}$ and $CR_{v \rightarrow i}$ to achieve the domain alignment, and design the dynamic interaction fusion that computes feature correlation matrix between two different-level tasks to achieve naturally semantic compatibility.

mismatch between at different-level tasks may lead to the vital information loss. On the one hand, unsupervised-based methods [14], [24] and self-supervised-based methods [25], [26] use the source images to supervise infrared encoder and visible encoder, which brings difficulty to extract domain alignment features. On the other hand, single-task fusion methods adopt feature fusion module (e.g., attentional fusion [23], [27], feature space transformation [17], [18], etc.) to achieve cross-modal feature fusion, but the fusion features lack object semantic information. Besides, multi-task fusion methods design different embedding ways (e.g., hierarchical interaction [1], bi-level formulation [2], progressive cycle attention [3], etc.) to embed object semantic information into the fusion network, but the feature maps between two different-level tasks have semantic mismatch problem. Moreover, hand-crafted losses (e.g., content loss [25], [28], structural loss [29], adversarial learning loss [18], [30], and joint multi-task loss [20], [21]), may lead to subjective preferences in the fusion images.

Based on the above analysis, since IVIF lacks ground truth, researchers empirically design the hand-crafted fusion losses that force the fused images with to obtain certain features (e.g., brightness, contrast, etc.), and thus these methods cannot comprehensively retain all the vital information from source images. Therefore, we propose an infrared and visible image fusion via cross reconstruction learning (CRL) that doesn't using any hand-crafted fusion losses, which is named as FreeFusion, as shown in Fig. 1(b). FreeFusion is not constrained by any predefined fusion rules, which enables flexibly fusing all vital features from source images. Specifically, CRL represents the fusion features by learning the inverse mapping knowledge,

where the infrared image is used as input and the visible image is reconstructed ($CR_{i \rightarrow v}$), and vice versa ($CR_{v \rightarrow i}$). If the fusion features fail to contain vital features of two modalities during the cross-reconstruction process, they cannot reconstruct another source image. Thus, without using any fusion loss, the fusion model can adaptively fuse complementary features of two modalities. Meanwhile, the infrared and visible features need to keep consistency in a same domain space, thereby achieving cross-modal feature domain alignment. Since the fusion and segmentation are two different-level tasks, the fusion features cannot naturally match object semantic features. Consequently, we introduce a dynamic interaction fusion (DIF_j, $j=1,2,3$) to achieve semantic compatibility and enhance fine-grained interactions between different task features. A correlation matrix between fusion features and object semantic features is built to overcome the semantic mismatch, and then the strong correlation features are enhanced and the weak correlation features are suppressed to improve the interactive ability. In the training phase, we share the decoder parameters of $CR_{i \rightarrow v}$, $CR_{v \rightarrow i}$, $SR_{v \rightarrow v}$ (self-reconstruction from visible image to visible image) and $SR_{i \rightarrow i}$ (self-reconstruction from infrared image to infrared image), which improves the robustness of the decoder to reconstruct the fusion image.

Structurally, a feature fusion module Φ_{FFM} generates fusion features, and then a feature transform module Φ_{FTM} transforms them into object semantic features. The feature maps between different-level tasks are embedded together through Φ_{CTIM} (CTIM denotes cross-task interaction module) based on the mixed correlation matrix, thus achieving object semantic information compatible with pixel-level fusion information. Next, a task query module Φ_{TQM} is designed to construct a query-based matrix multiplication that establishes dependency relationships between feature maps, thereby refining the spatial fine-grained information. Then, a feature decoupling module Φ_{FDM} decouples the fusion features to another modality features. Finally, the modality features and decoupled modality features are fed into the self-reconstruction decoder Φ_D^{sr} and cross reconstruction decoder Φ_D^{cr} to reconstruct the source images and another modality image. In the inference stage, we generate a fusion image by directly inputting the fusion features to the decoder. As shown in Fig. 2(Right), compared to the second best methods, our FreeFusion improves 0.8%, 2.78%, 2.27%, 4.01%, 0.98%, 2.35% of EN, SD, SF, AG, SCD, and VIF on Potsdam dataset, respectively.

In summary, our contributions are as follows:

- We propose a cross-reconstruction learning method, which eliminates the reliance on hand-crafted fusion losses in infrared and visible image fusion. The complementary information of source images is fused more efficiently and adaptively, facilitating the fusion and downstream tasks.
- CRL is built that decouples the fusion features to reconstruct another-modality source image, which forces the infrared and visible features to be coincident in a same domain space.
- DIF_j is designed through building a correlation matrix between fusion features and object semantic features, and

Methods	YDTR	Tardal	U2Fusion	PIAFusion	SegMif	CDDFuse	Ours
L_1	✗	✓	✗	✓	✗	✗	✗
L_2	✓	✗	✓	✗	✓	✓	✗
$SSIM$	✓	✓	✓	✗	✓	✓	✗
SF	✓	✗	✗	✗	✗	✗	✗
Aux	✗	✗	✗	✓	✗	✗	✗
$Texture$	✗	✗	✗	✓	✗	✗	✗
$Grad$	✗	✗	✗	✗	✓	✓	✗
$Decomp$	✗	✗	✗	✗	✗	✓	✗

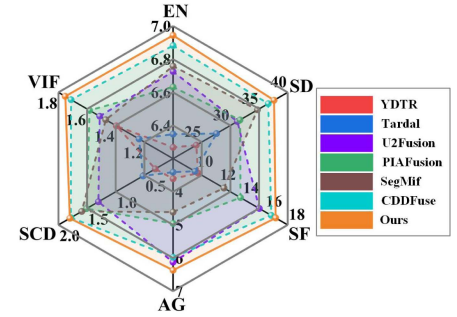


Fig. 2. Left: Illustration of hand-crafted fusion losses constructed by different methods. ✓ indicates the fusion loss is used, and ✗ indicates the fusion loss isn't used. Right: Quantitative comparison of our method and the state-of-the-art methods on Potsdam dataset.

then improves their interactive ability by enhancing the strong correlation features.

II. RELATED WORK

A. Unsupervised Fusion Methods

Unsupervised fusion methods focus on constructing hand-crafted fusion losses that push the fusion image to resemble the source images. For example, Zhao et al. [16] propose a correlation-driven feature decomposition fusion method, and design a correlation-driven loss to retain more modality-shared and modality-specific information. Tang et al. [15] design a progressive image fusion network that adaptively integrates meaningful information by constructing an illumination-aware loss. Zhao et al. [25] explore an interactive feature embedding method that adopts structural similarity loss to retain the important information of source images. Ma et al. [18] construct dual-discriminator conditional generative adversarial network, which designs a content loss to constrain the pixel intensity and gradient variation between fusion image and source images. Rao et al. [19] adopt a lightweight transformer module and adversarial learning algorithm to construct variance-based structural similarity loss that constrains the fusion image. Yao et al. [31] propose a color image fusion framework, where a gaussian blur kernel-based color loss is constructed to retain color information from visible images under low-light conditions.

However, those fusion methods construct hand-crafted fusion losses, which may not retain all the helpful information of source images. In contrast, we propose the cross reconstruction learning that doesn't use any hand-crafted fusion losses. Since the fusion results are not subject to the assumption of any specific prior knowledge, the fusion model can freely fuse more comprehensive information.

B. Self-Supervised Fusion Methods

Self-supervised fusion methods aim to train encoder-decoder models for feature extraction and image reconstruction, and design specific fusion strategies to preserve the specific features of source images. For instance, Li et al. [23] develop a nest-connection method, which utilizes the multi-scale deep feature fusion strategy to preserve more salient features. Zhao et al. [22] explore deep image decomposition strategy, where loss

function is designed to make the background/detail feature maps of the source images similar/dissimilar. Li et al. [29] devise a residual fusion network based on the residual architecture, and propose a detail preservation loss and feature enhancement loss to force the network to fuse more detail features. Liang et al. [32] design an image decomposition model by self-supervised representation learning, which can achieve image fusion without any paired data or sophisticated data loss. Tang et al. [33] utilize scene illumination disentangled network to reconstruct two source images while preserving the useful information of source images. Xu et al. [34] decompose source images into scene features and attribute vectors by a cross-modal reconstruction. Then, the scene features and attribute vectors of the two modalities are fused using hand-crafted averaging and addition strategies, respectively. However, hand-crafted fusion strategies lack adaptivity, resulting in the loss of vital features. In contrast, we propose the cross-reconstruction learning to decouple the fusion feature into two modal features that crossly reconstruct the source images. Thus, the cross-reconstruction process forces the fusion feature to adaptively contain all the vital information from source images.

The above fusion methods perform end-to-end training using the source images as ground truth, but it is difficult to adequately address the domain discrepancy between source images, resulting in some important detail information loss. In contrast, our cross reconstruction learning decouples the fusion features to reconstruct another-modality source image. During the decoupling and reconstruction process, the visible and infrared features are forced to keep consistency in a same domain space, thus achieving the cross-modal feature domain alignment.

C. Joint Multi-Task Fusion Methods

Recently, joint multi-task fusion methods attempt to strengthen the performance of fusion networks using semantic information from high-level vision tasks. They are divided into two classes: cascade learning methods [1], [35], [36] and interactive learning methods [2], [20], [21], [37]. Cascade learning methods train the fusion network by using high-level tasks as constraints, which prompts the fusion network to generate fusion results that benefit downstream tasks. For example, Sun et al. [35] obtain object-related information through a object detection network and then use the object location as a priori

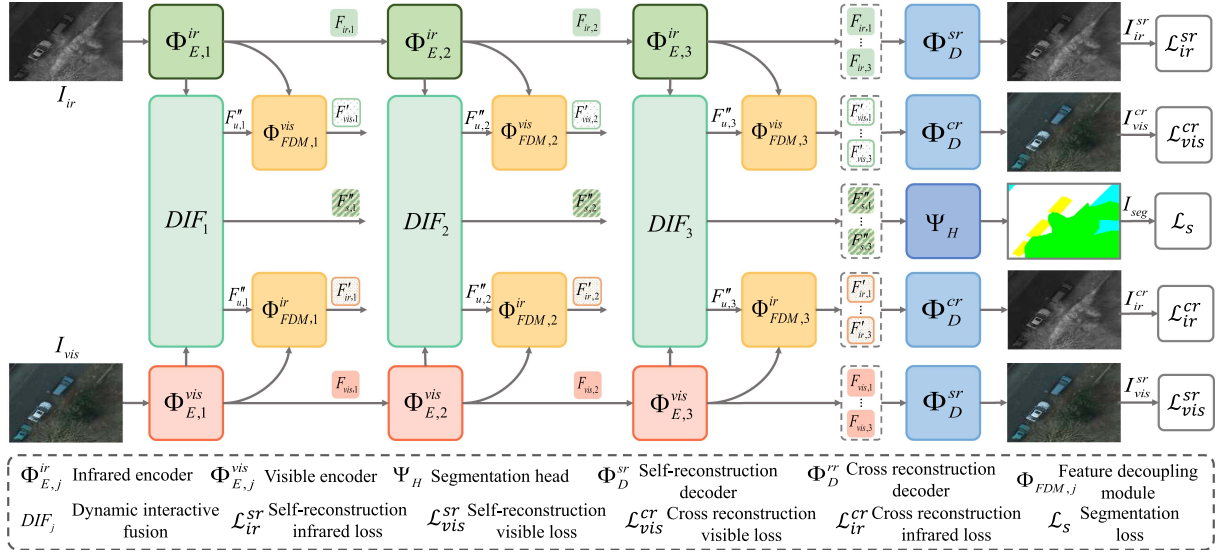


Fig. 3. Overall framework of the proposed FreeFusion. Without using any hand-crafted fusion loss, FreeFusion designs a cross-reconstruction learning (CRL) framework that learns the inverse mapping knowledge to represent fusion features. Firstly, $\Phi_{E,j}^{ir}$ and $\Phi_{E,j}^{vis}$ extract infrared and visible features. Then, DIF_j builds a correlation matrix between fusion features and object semantic features to achieve semantic compatibility. Next, $\Phi_{FDM,j}^{vis}$ and $\Phi_{FDM,j}^{ir}$ decouple the fusion features to reconstruct another-modality source image, respectively. The decoder parameters of Φ_D^{sr} and Φ_D^{cr} are shared that improves the robustness of the decoder. In the inference stage, the fusion features are fed directly into the decoder to generate a fusion image.

information to guide the fusion network. Liu et al. [1] combine dual adversarial learning and detection networks and propose a two-layer optimization formulation to strengthen the link between the fusion and segmentation tasks. Tang et al. [36] utilize a semantic loss to guide the flow of high-level semantic information back to the image fusion module, alleviating the gap between the fusion and segmentation tasks. Wu et al. [38] use a warm-start fusion loss to obtain initial fusion results, and then directly introduces a segmentation loss to perform semantic-driven training. Yao et al. [39] implement an intensity and gradient-based fusion loss to train a laplacian pyramid fusion network. Then, the fusion result is appended to a segmentation network which is trained with a cross-entropy loss. Thus, both semantic-level loss and fusion-level loss are used in the two methods. In contrast, our method doesn't employ any fusion loss, i.e., we decouple the fusion feature by crossly reconstructing the source images, which ensures the model freely fuses the complementary features of two modals. Interactive learning methods design hierarchical interactions between feature maps of different levels, thereby bridging the gap between the fusion and high-level tasks. For example, Zhao et al. [20] address the discrepancy between semantic features and fusion features through a meta-feature embedding for object detection. Wang et al. [37] explore the collaborative relationship between fusion and salient object detection tasks, and design an interactive enhanced multi-task paradigm. Liu et al. [21] propose a multi-interaction feature learning architecture that utilizes dual-task correlation to improve the performance of dual tasks. Moreover, some studies combine low-level vision tasks to improve the performance of fusion task. For example, Jie et al. [40] combine the super-resolution task to implement a random iterative denoising process, thereby generating high-resolution fusion results. Li et al. [41] first attempt to address the image registration and

fusion problem in a single-stage framework, which facilitates mutual enhancement of the two tasks. Li et al. [42] utilize the registration network to obtain alignment knowledge that supervises mismatched image fusion, freeing the fusion algorithm from strict registration. Li et al. [43] jointly train image fusion and restoration tasks, which can effectively fuse and recover the degraded multi-modality information.

Existing methods embed the object semantic information from the high-level visual task into the fusion task, but two different-level tasks hinder this process. Thus, we propose the dynamic interaction fusion to achieve semantic compatibility and refine the feature representation of each task. Especially, we leverage the correlation matrix and query-level interactions, which enables the object semantic information to be naturally compatible to the fusion task.

III. PROPOSED METHOD

Since no ground truth can be used to train the fusion models, existing methods construct elaborate fusion losses that may not retain all the vital features of source images. As is illustrated in Fig. 3, FreeFusion aims to design a CRL framework that doesn't use any hand-crafted fusion losses, while forcing the network to adaptively fuse complementary information of source images. Especially, we design DIF_j to achieve naturally semantic compatibility between fusion and segmentation tasks. Details are introduced as follows.

A. Cross Reconstruction Learning

Without using any hand-crafted fusion loss to train the fusion model, which brings the challenges of how to force the network to freely extract and fuse all the vital features and then reconstruct the fusion image. Our method learns the inverse mapping

knowledge by implementing $CR_{i \rightarrow v}$ and $CR_{v \rightarrow i}$, which forces the fusion features to contain key features of the two modalities.

Feature extraction via $SR_{i \rightarrow i}$ and $SR_{v \rightarrow v}$: We extract the vital features from source images by decoupling the fusion features to restructure the corresponding source image. Firstly, given input infrared image I_{ir} and visible image I_{vis} , the infrared and visible encoders are represented as $\Phi_{E,j}^{ir}$ and $\Phi_{E,j}^{vis}$, where $j = 1, 2, 3$ is the number of layers. In each layer, two modality encoders perform feature extraction through $\Phi_{E,j}^{ir}$ and $\Phi_{E,j}^{vis}$ to generate infrared feature $F_{ir,j}$ and visible feature $F_{vis,j}$. Then, infrared and visible features from each layer are fed to the self-reconstructing decoder Φ_D^{sr} , which reconstructs the infrared image I_{ir}^{sr} and visible image I_{vis}^{sr} . The infrared image reconstruction loss $\mathcal{L}_{ir}^{sr}$ and visible image reconstruction loss $\mathcal{L}_{vis}^{sr}$ ensure that the learned comprehensive features of source images are not lost during the self-reconstruction process, which can be written as:

$$\mathcal{L}_{ir}^{sr} = \|I_{ir} - I_{ir}^{sr}\|_1 + 1 - \mathcal{S}(I_{ir}, I_{ir}^{sr}), \quad (1)$$

$$\mathcal{L}_{vis}^{sr} = \|I_{vis} - I_{vis}^{sr}\|_1 + 1 - \mathcal{S}(I_{vis}, I_{vis}^{sr}), \quad (2)$$

where $\mathcal{S}(\cdot, \cdot)$ denotes structural similarity index [25].

Feature fusion via $CR_{i \rightarrow v}$ and $CR_{v \rightarrow i}$: If the fusion features can be decoupled that generates the cross-modality images, it will be forced to freely fuse the important feature information of two modalities. Meanwhile, the infrared and visible features have to be domain alignment in a same domain space. As shown in Fig. 3, the fusion-related feature $F''_{u,j}$ from DIF_j (details in Section III-B) and the infrared feature $F_{ir,j}$ are fed into $\Phi_{FDM,j}^{vis}$ to generate the decoupled visible feature $F'_{vis,j}$. Similarly, the decoupled infrared feature $F'_{ir,j}$ can be obtained via $\Phi_{FDM,j}^{ir}$. Then, the decoupled infrared and visible features from each layer are input to the cross reconstruction decoder Φ_D^{cr} that generates the cross-reconstructed infrared image I_{ir}^{cr} and the cross-reconstructed visible image I_{vis}^{cr} , respectively. The cross reconstruction losses $\mathcal{L}_{ir}^{cr}$ and $\mathcal{L}_{vis}^{cr}$ mainly ensure that the information of two modalities contained in the fusion features is not lost. They can be written as:

$$\mathcal{L}_{ir}^{cr} = \|I_{ir} - I_{ir}^{cr}\|_1 + 1 - \mathcal{S}(I_{ir}, I_{ir}^{cr}), \quad (3)$$

$$\mathcal{L}_{vis}^{cr} = \|I_{vis} - I_{vis}^{cr}\|_1 + 1 - \mathcal{S}(I_{vis}, I_{vis}^{cr}). \quad (4)$$

Finally, $SR_{i \rightarrow i}$, $SR_{v \rightarrow v}$, $CR_{i \rightarrow v}$ and $CR_{v \rightarrow i}$ train the four shared-parameter decoders, thus enabling the decoder to achieve the generalization ability of the image reconstruction. Therefore, in the inference phase, the fusion features are fed directly to the decoder that can generate a fusion image.

B. Dynamic Interactive Fusion

Fusion features can help segmentation task to provide pixel-level detail information, and object semantic features can guide fusion task to contain rich semantic information. But fusion and segmentation tasks have semantic mismatch due to their task levels' difference. Thus, as shown in Fig. 4, we design DIF_j to achieve semantic compatibility between two different-level tasks. In this subsection, we emphasize two stages: cross-task

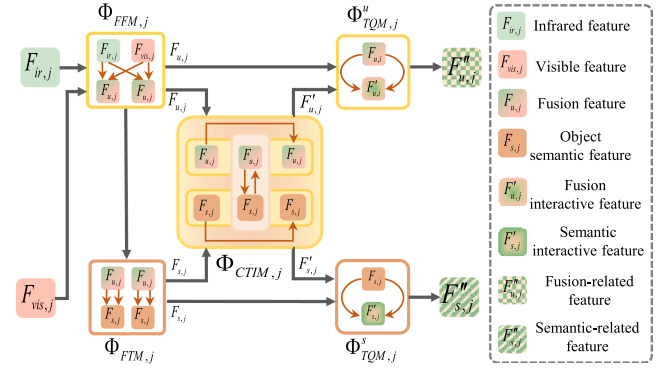


Fig. 4. Illustration of dynamic interactive fusion DIF_j . DIF_j consists of two stages: cross-task interaction and task querying. In the cross-task interaction, $\Phi_{CTIM,j}$ utilizes the correlation matrix that performs elemental multiplication to achieve semantic compatibility between $F_{u,j}$ and $F_{s,j}$, thereby facilitating intra-task self-reinforcement and inter-task complementary abilities. In the task query, $F_{u,j}$ and $F_{s,j}$ are fed into $\Phi_{TQM,j}^u$ to refine spatial fine-grained information of the fusion task through query-level global interaction.

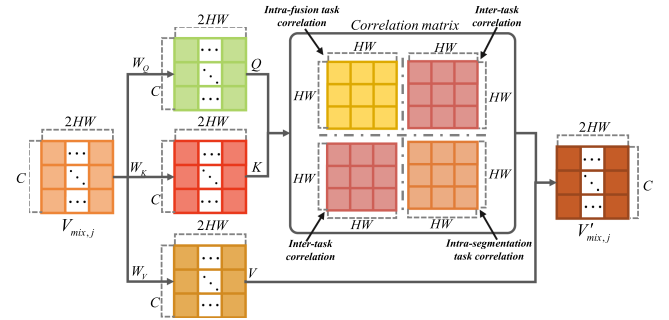


Fig. 5. Illustration of cross-task interaction module Φ_{CTIM} . The mix vector $V_{mix,j}$ is spatially interacted by matrix multiplication in Q and K to generate the correlation matrix, which achieves the semantic compatibility and enhances intra-task and inter-task feature representations.

interaction and task query, which will be introduced in details as follows.

In the cross-task interaction stage, we introduce $\Phi_{CTIM,j}$ that establishes spatially fine-grained interaction between intra task and inter task to strengthen the mutually beneficial feature representations, as shown in Fig. 5. We utilize a correlation matrix to weight each pixel position information of the input feature maps, thereby enhancing the potential connection between fusion and semantic features. We can naturally infer four blocks of correlation matrices: intra-fusion task correlation, intra-segmentation task correlation, and inter-task correlation between two tasks.

Specifically, in each layer, we feed $F_{vis,j}$ and $F_{ir,j}$ into $\Phi_{FFM,j}$ to generate fusion feature $F_{u,j}$, and then the fusion feature is transformed into the object semantic feature $F_{s,j}$ by $\Phi_{FTM,j}$. Next, $F_{u,j}$ and $F_{s,j}$ are flattened into fusion vector $V_{u,j}$ and object semantic vector $V_{s,j}$. We concatenate $V_{u,j}$ and $V_{s,j}$ to get mix vector $V_{mix,j}$. $V_{mix,j}$ is multiplied by the learnable weight matrices W_Q , W_K , and W_V to obtain Q , K and V respectively, which can be expressed as follows:

$$\{Q, K, V\} = \{V_{mix,j}W_Q, V_{mix,j}W_K, V_{mix,j}W_V\}. \quad (5)$$

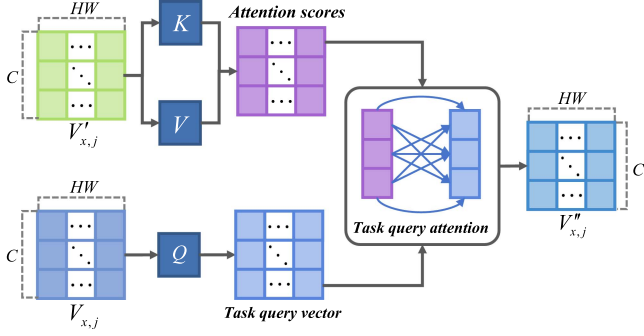


Fig. 6. Illustration of task query module $\Phi_{TQM,j}^u$ or $\Phi_{TQM,j}^s$. Task interaction vector $V'_{x,j}$ as K and V generates attention scores by matrix multiplication. The task query vector $V_{x,j}$ as Q is then modulated by the attention scores to refine the spatial fine-grained information.

Subsequently, the correlation matrix is computed using the scaled dot product between Q and K , and then normalize them into $[0, 1.0]$ with softmax function. After that, the correlation matrix is multiplied with V to enhance mix vector representation between the intra task and inter task. To capture different contextual information in $V_{mix,j}$, multiple attention heads are computed in parallel. This mechanism facilitates global interaction and fusion for fusion and segmentation tasks. Obviously, we achieve semantic compatibility between fusion and semantic features by increasing the weight values of strong semantic correlation positions and decreasing the weight values of weak semantic correlation positions. The value vector is weighted by correlation matrix to enhance intra-task itself information and inter-task complementary information. The attention mechanism is calculated as follows:

$$V'_{mix,j} = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad (6)$$

where $\sqrt{d_k}$ is the scaling factor, and $V'_{mix,j}$ denotes the output refined mixing vector. Finally, we split $V'_{mix,j}$ according to $\text{reshape}(\text{split}(V'_{mix,j}))$, thereby generating fusion interactive feature $F'_{u,j}$ and semantic interactive feature $F'_{s,j}$.

In the task query stage, we introduce two task query modules (i.e., $\Phi_{TQM,j}^u$ and $\Phi_{TQM,j}^s$) that construct spatially fine-grained interactions to generate more refined task-specific feature representations. We denote all subscripts uniformly as $x = \{u, s\}$ to simplify the expression, where $F_{u,j}$ and $F_{s,j}$ are defined as task query feature $F_{x,j}$, and $F'_{u,j}$ and $F'_{s,j}$ are defined as task interactive feature $F'_{x,j}$. Specifically, $F_{x,j}$ and $F'_{x,j}$ are transformed into the task query vector $V_{x,j}$ and task interactive vector $V'_{x,j}$. As shown in Fig. 6, different from the original self-attention mechanism, we take $V_{x,j}$ as task query and $V'_{x,j}$ as key and value. Thus, $V_{x,j}$ and the learned $V'_{x,j}$ construct long-range interactions through a matrix multiplication, resulting in fine-grained task-specific feature representation. $V_{x,j}$ and $V'_{x,j}$ are mapped to $\hat{Q}_x$, $\hat{K}_x$ and $\hat{V}_x$ through three learnable matrices $W_{\hat{Q}}$, $W_{\hat{K}}$, and $W_{\hat{V}}$, which can be formulated as:

$$\{\hat{Q}_x, \hat{K}_x, \hat{V}_x\} = \{V_{x,j}W_{\hat{Q}}, V'_{x,j}W_{\hat{K}}, V'_{x,j}W_{\hat{V}}\}. \quad (7)$$

Here, $\hat{K}_x$ and $\hat{V}_x$ are performed the scaled dot product computation, and then the attention scores are obtained by *softmax* operation. Next, attention scores are applied to $V_{x,j}$ to generate refined task-related vector $V'_{x,j}$. Finally, $V'_{x,j}$ generates task-related feature $F'_{x,j}$ through *reshape* operation, where $F'_{x,j}$ denotes fusion-related feature $F'_{u,j}$ and semantic-related feature $F'_{s,j}$, respectively.

C. Architecture

As shown in Fig. 3, our method is composed by CRL framework, which includes $\Phi_{E,j}^{ir}$, $\Phi_{E,j}^{vis}$, $\Phi_{FDM,j}^{vis}$, $\Phi_{FDM,j}^{ir}$, Φ_D^{sr} , Φ_D^{cr} and Ψ_H . $j = 1, 2, 3$ denotes the feature level index.

Specifically, $\Phi_{E,j}^{ir}$ includes three encoder blocks, where each encoder block contains one convolution layer with 3×3 kernel and one residual layer. The residual layer consists of two convolution layers with 3×3 kernels and skip connection. $\Phi_{E,j}^{vis}$ has the same structure. Φ_D^{sr} includes three decoder blocks. Each decoder block consists of one convolution layer with 3×3 kernel, one residual block, and one up-sampling. The up-sampling conducts bilinear interpolation and one convolution layer with 1×1 kernel. Ψ_H is divided into three blocks. The first two blocks have the same structure as the decoder block. The last block consists of one convolution layer with 3×3 kernel, one residual block and one classification layer. The classification layer is one convolutional layer with 1×1 kernel. In addition, we add skip connections between encoder-decoder to fully fuse multi-scale features. The skip connections are also executed between DIF_j and Ψ_H . $\Phi_{FDM,j}$ is used to decouple the fusion features to generate another two modality features. $\Phi_{FDM,j}$ includes $\Phi_{FDM,j}^{vis}$ and $\Phi_{FDM,j}^{ir}$. To generate the decoupled visible feature $F'_{vis,j}$, we feed the fusion-related feature $F'_{u,j}$ and the infrared feature $F_{ir,j}$ into $\Phi_{FDM,j}^{vis}$. Firstly, $F_{ir,j}$ is fed into a nonlinear transform consisting of one convolution layer with 1×1 kernel and PReLU and one convolution layer with 1×1 kernel and Sigmoid to generate the enhancement coefficients. Then, $F_{ir,j}$ and the enhancement coefficients are multiplied to obtain infrared feature that emphasizes the salient part of itself. Finally, $F'_{u,j}$ and the enhanced infrared feature perform element-wise subtraction to generate $F'_{vis,j}$. The whole process can be represented as:

$$F'_{vis,j} = F_{u,j} - (F_{ir,j} \otimes \sigma(C_1^1(\delta(C_1^1(F_{ir,j}))))), \quad (8)$$

where σ and δ denote Sigmoid and PReLU activation functions, respectively. $\otimes$ denotes element-wise multiplication. Besides, $\Phi_{FDM,j}^{ir}$ generates the decoupled infrared feature $F'_{ir,j}$, which has the same operation.

DIF_j includes $\Phi_{FFM,j}$, $\Phi_{FTM,j}$, $\Phi_{CTIM,j}$ and $\Phi_{TQM,j}$. The detailed structures of $\Phi_{CTIM,j}$ and $\Phi_{TQM,j}$ are provided in Section III-B. $\Phi_{FFM,j}$ fuses infrared and visible features to generate fusion features. We first concatenate modality features $F_{ir,j}$, $F_{vis,j}$, and then obtain fusion features by conducting three convolution layers with 3×3 kernel. Next, we obtain the weight maps W_1 , W_2 from infrared and visible images by conducting one convolution layers with 1×1 kernel. The weight map W_1

is formulated as:

$$W_1 = C_1^1 \left(C_3^3 \left(F_{ir,j} \biguplus F_{vis,j} \right) \right), \quad (9)$$

where C_k^n denotes that the number of "Convolution with $k \times k$ kernel+ReLU" layers is n , and $\biguplus$ denotes feature concatenation. Finally, the enhanced modality features are generated by using channel multiplication for $F_{ir,j}$ and $F_{vis,j}$. W_2 is obtained in the same way. Then, we concatenate these two features and fusion feature $F_{u,j}$ by conducting once convolution layer with 3×3 kernel. The feature fusion process is denoted as:

$$F_{u,j} = C_3^1 \left(F_{vis,j} W_1 \biguplus F_{ir,j} W_2 \right). \quad (10)$$

$\Phi_{FTM,j}$ transforms the fusion feature into semantic feature. We perform one convolution layer with 3×3 kernel and one convolution layer with 1×1 kernel on the fusion feature. Then, it is concatenated with the fusion features to generate semantic features, which can be expressed as:

$$F_{s,j} = C_1^1 \left(C_3^1 (F_{u,j}) \right) + F_{u,j}. \quad (11)$$

D. Model Training and Inference

Training: We jointly train $SR_{i \rightarrow i}$, $SR_{v \rightarrow v}$, $CR_{i \rightarrow v}$, $CR_{v \rightarrow i}$ and segmentation task. Thus, the total loss function can be formulated as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{ir}^{sr} + \mathcal{L}_{vis}^{sr} + \mathcal{L}_{ir}^{cr} + \mathcal{L}_{vis}^{cr} + \alpha \mathcal{L}_s, \quad (12)$$

where α is a hyperparameter to balance the fusion and segmentation tasks. $\mathcal{L}_s = \mathcal{L}_{ce} + \mathcal{L}_{dice}$ denotes segmentation loss, which contains a cross-entropy loss $\mathcal{L}_{ce}$ and dice loss $\mathcal{L}_{dice}$.

If we consider the classification probability of pixels, each pixel is guaranteed to be classified accurately. $\mathcal{L}_{ce}$ provides accuracy in category prediction, which is expressed as follows:

$$\mathcal{L}_{ce} = -\frac{1}{Y} \sum_{y=1}^Y \sum_{c=1}^C g_{y,c} \log(p_{y,c}), \quad (13)$$

where C and Y denote the number of classes and pixels, respectively. $g_{y,c}$ represents the probability of the y -th pixel belonging to class c , and $p_{y,c}$ is the corresponding semantic label. $\mathcal{L}_{dice}$ ensures that small or unbalanced categories are handled appropriately by optimizing the overall regional similarity, which is expressed as follows:

$$\mathcal{L}_{dice} = 1 - \frac{1}{C} \sum_{c=1}^C \frac{2 \sum_{y=1}^Y p_{y,c} g_{y,c} + \varepsilon}{\sum_{y=1}^Y p_{y,c} + \sum_{y=1}^Y g_{y,c} + \varepsilon}, \quad (14)$$

where ε is set to 1×10^{-7} to ensure that the denominator is not zero.

Inference: The overall framework relies on the information of the source images, which ensures that Φ_D^{cr} and Φ_D^{sr} contain comprehensive vital features of source images. Since Φ_D^{cr} and Φ_D^{sr} share parameters, we uniformly denote as decoder Φ_D . Thus, the fusion results can be generated by Φ_D in the inference stage, which can be written as:

$$I_f = \Phi_D (F_{u,1}''', F_{u,2}''', F_{u,3}'''), \quad (15)$$

where $F_{u,j}''$, $j = 1, 2, 3$ denotes the fusion-related features generated by $\Phi_{FTM,j}''$, and I_f denotes the fusion result.

IV. EXPERIMENTS

A. Setup

Datasets: We conduct experiments on five representative datasets: Potsdam [44], WHU [45], MFNet [46], M³FD [1] and LLVIP [47]. The Potsdam dataset provides detailed information about the urban environment, including 6 categories of objects: *Impervious surfaces, Building, Low vegetation, Tree, Car and Clutter*. It is divided into 30 image pairs for training and 7 image pairs for testing. The WHU dataset describes the scenario of land, which is labeled as 7 different categories including *Farmland, City, Village, Water, Forest, Road, and Others*. It includes 100 image pairs with 80 image pairs for training and 20 image pairs for testing. The MFNet dataset presents an urban scene about daytime and nighttime, which has 9 different object categories, including *Background, Car, Person, Bike, Curve, Car_stop, Guardrail, Color_cone, and Bump*. It contains 1177 image pairs with 784 image pairs for training and 393 image pairs for testing. Besides, three datasets are used to evaluate the performance of the semantic segmentation task. M³FD dataset covers different scenarios of the illumination, season and weather, where six object categories are annotated, i.e., *People, Car, Bus, Motorcycle, Truck and Lamp*. LLVIP dataset provides a road pedestrian scene at night and daylight, which is labeled with a pedestrian category. M³FD with 1260 image pairs and LLVIP with 3463 image pairs are used for testing to evaluate fusion performance. In addition, M³FD and LLVIP are divided into training set (2940 and 12025 images) and test set (1260 and 3463 images) to evaluate the performance of object detection task, respectively.

Implementation: Our FreeFusion is implemented with pytorch framework on NVIDIA GeForce RTX 4090 GPU. The learning rate is initialized to 2×10^{-5} and gradually decays to 1×10^{-6} . The batch size is taken to 4. We use Adam optimizer with first-order momentum 0.9 and second-order momentum 0.99 to train FreeFusion for 50 epochs. All images are cropped to the size of 320×320 .

Metric: Six metrics are used to quantitatively measure our method: entropy (EN) [48], standard deviation (SD) [49], spatial frequency (SF) [50], average gradient (AG) [51], structural content difference (SCD) [49], and visual fidelity (VIF) [52]. A larger EN means that the fusion image contains richer details and information. A higher SD means that the image has higher contrast and brightness variations. A larger SF indicates richer texture representation in the fusion image. A higher AG indicates sharper edges contained in the fusion result. A larger SCD indicates that the fusion image better preserves the structural information of source images. A higher VIF indicates that the fusion image is better perceived visually by the human eyes. Besides, we use mDice and mAcc metrics to evaluate the performance of the segmentation task. mDice is higher that means the model has better segmentation performance on all categories. A higher mAcc indicates that the model classifies pixels more accurately. We use mAP_{50→95} to evaluate the performance of

TABLE I
EFFECTIVENESS VALIDATION OF CROSS-TASK INTERACTIVE MODULE (Φ_{CTIM}) ON POTSDAM DATASET

Method	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$	mDice $\uparrow$	mAcc $\uparrow$
Concatenate	6.805	34.382	10.034	3.988	1.440	1.231	72.04	69.75
Summary	6.864	35.384	9.902	3.983	1.438	1.762	73.32	72.05
w/o Φ_{CTIM}	6.798	35.276	12.231	4.812	1.569	1.395	72.53	70.95
w/ Φ_{CTIM}	6.944	37.616	16.946	6.370	1.800	1.752	75.31	74.02

The best result is in red and the second best one is in blue.

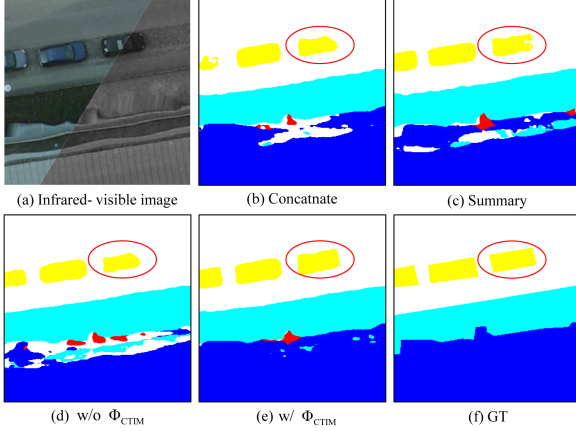


Fig. 7. Visual comparison of segmentation results produced by Concatenate, Summary, w/o Φ_{CTIM} and w/ Φ_{CTIM} on Potsdam dataset.

object detection task. $mAP_{50 \rightarrow 95}$ denotes the average value of AP at different IoU thresholds of 0.5 to 0.95 by an interval of 0.05. Higher $mAP_{50 \rightarrow 95}$ indicate a better ability to localize objects.

B. Ablation Study

Analysis of the cross-task interaction module Φ_{CTIM} : Since fusion and segmentation are two different-level tasks, there are semantic mismatch between them. As described in Section II-I-B, we introduce Φ_{CTIM} to implement semantic alignment for pixel-level fusion features and object semantic features. To verify its effect, we study the performance impact of different interaction ways on fusion results and semantic segmentation. As shown in Table I, our FreeFusion (w/ Φ_{CTIM}) achieves the best results. Simple aggregation operations (Concatenate and Summary) cannot directly fuse two features with semantic mismatch, resulting in the fusion features containing less information (e.g., pixel intensity and gradient). Moreover, the method w/o Φ_{CTIM} directly utilizes segmentation features to prompt fusion features through the back-propagation manner, which has limited guidance for fusion performance. In contrast, the method w/ Φ_{CTIM} obtains the best metric scores, improving EN/SD/SF/AG/SCD/VIF by 2.3% / 6.6% / 38.5% / 32.4% / 14.7% / 25.6% relative percentage than w/o Φ_{CTIM} . Through qualitative and quantitative segmentation experiment results, we find that different variants (Concatenate, Summary, and w/o Φ_{CTIM}) cause segmentation performance degradation in different degrees. w/ Φ_{CTIM} outperforms the second best method 2.7% and 2.7% on mAcc and mDice. As shown in Fig. 7(b-d), the contours of object (e.g., car) cannot be precisely segmented,

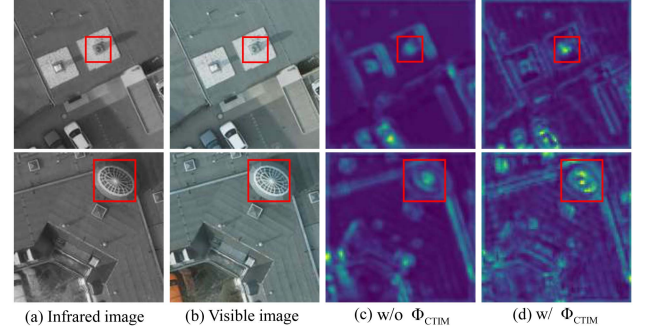


Fig. 8. Feature visualization of w/o Φ_{CTIM} and w/ Φ_{CTIM} . (c) denotes the fusion feature $F_{u,2}$ is obtained from $\Phi_{FFM,2}$ output, and (d) denotes the fusion interactive feature $F'_{u,2}$ is obtained from $\Phi_{CTIM,2}$ output.

TABLE II
IMPORTANCE STUDY OF TASK QUERY MODULE (Φ_{TQM}) ON POTSDAM DATASET

Method	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$
w/ Φ_{TQM}^u	6.778	33.325	10.986	4.367	1.403	1.339
w/ Φ_{TQM}^s	6.504	28.067	13.768	5.325	0.622	1.287
w/o Φ_{TQM}	6.578	29.057	12.922	5.149	0.951	1.264
w/ Φ_{TQM}	6.944	37.616	16.946	6.370	1.800	1.752

The best result is in red and the second best one is in blue.

which illustrates the fusion results generated by these variants fail to preserve the object edges. The reason is that they ignore the feature mismatch of different-level tasks, thereby limiting the effect of the semantic feature integration. As shown in Fig. 8, we visualize representative features to discuss the effectiveness of Φ_{CTIM} in DIF_2 . Obviously, the fusion feature lacks object semantic information (e.g., roof objects) in Fig. 8(c), resulting in failure to adequately extract semantic-level feature representations. In contrast, as shown in Fig. 8(d), w/ Φ_{CTIM} achieves semantic compatibility between two different-level tasks to promote fusion feature containing more object semantic information, resulting in a better fusion performance.

Effectiveness of task query module Φ_{TQM} : In Section III-B, we introduce Φ_{TQM} to refine the task-related feature maps. To validate the effectiveness of Φ_{TQM} in our FreeFusion, we adopt w/ Φ_{TQM}^u (FreeFusion with Φ_{TQM} in CRL), w/ Φ_{TQM}^s (FreeFusion with Φ_{TQM} in the segmentation task), w/o Φ_{TQM} (FreeFusion without Φ_{TQM} in two tasks) and w/ Φ_{TQM} (FreeFusion), for comparison. Table II shows quantitative results of different methods. w/ Φ_{TQM} obtains the best fusion performance compared to the other three variants. Moreover, w/ Φ_{TQM}^u achieves nice performance that outperforms w/ Φ_{TQM}^s by 4.2% / 18.7% / 126% / 4% relative percentage in EN/SD/SCD/VIF. w/ Φ_{TQM}^u is also superior to w/o Φ_{TQM} , raising 3% / 17% / 47.5% / 5.9% relative percentage in EN/SD/SCD/VIF. w/ Φ_{TQM}^u can improve fusion performance to some extent comparing with w/ Φ_{TQM}^s and w/o Φ_{TQM} , but w/ Φ_{TQM}^u is still lower than w/ Φ_{TQM}^u in EN/SD/SF/AG/SCD/VIF by 2.4% / 12.9% / 54.3% / 45.9% / 28.3% / 30.8% relative percentage. The quantitative results comparing the three variants demonstrate that Φ_{TQM} enhances the fine-grained representation of the intra-task feature maps, thus improving the fusion performance.

TABLE III
INFLUENCE STUDY OF FEATURE DECOUPLING MODULE (Φ_{FDM}) ON
POTSDAM DATASET

Method	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$	mDice $\uparrow$	mAcc $\uparrow$
w/o Φ_{FDM}	6.458	26.707	13.902	5.545	0.371	1.237	74.80	73.31
w/ Φ_{FDM}	6.944	37.616	16.946	6.370	1.800	1.752	75.31	74.02

The best result is in red and the second best one is in blue.

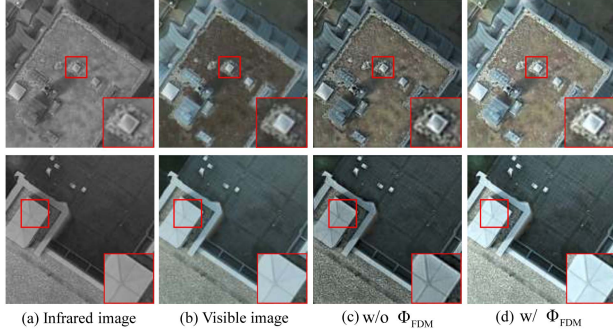


Fig. 9. Visual comparison of fusion results generated by w/o Φ_{FDM} and w/ Φ_{FDM} on Potsdam dataset.

Influence of feature decoupling module Φ_{FDM} : We design Φ_{FDM} to crossly reconstruct source images that forces the fusion features to retain the vital information from source images, and thus generating the fusion image without using any fusion loss. To validate the effectiveness of Φ_{FDM} , the method w/o Φ_{FDM} directly reconstructs the corresponding infrared image and visible image using the fusion features, rather than decoupling another modality features from the fusion features. As can be seen in Table III, the fusion performance of w/ Φ_{FDM} outperforms w/o Φ_{FDM} by 7.5%/40.8%/21.9%/14.9%/385.2%/41.6% relative percentage in EN, SD, SF, AG, SCD, VIF, respectively. The segmentation performance of w/ Φ_{FDM} outperforms w/o Φ_{FDM} by 0.68%/10.96% relative percentage in mDice and mAcc, respectively. Moreover, we provide the fusion results of w/o Φ_{FDM} and w/ Φ_{FDM} in Fig. 9. Although w/o Φ_{FDM} fails to fully preserve the main information from source images, some vital information (e.g., roof and its texture) cannot be effectively highlight. In contrast, w/ Φ_{FDM} highlights the texture details, which can obviously improve the fusion performance. Moreover, a single infrared or visible image cannot guarantee the accuracy of segmentation results in Fig. 10(b)–(d), e.g. building and car. w/o Φ_{FDM} improves the accuracy of segmentation results, but it still fails to recognize the categories about "impervious surfaces" and "Building". Qualitative and quantitative evaluations of w/o Φ_{FDM} demonstrate the importance of decoupling another modality features to crossly reconstruct the source images.

Analysis of different α : Table IV shows the impact of different α on the fusion performance. When $\alpha = 0.1$, the model cannot fully utilize the object semantic information to help fusion, which limits the fusion effect. When $\alpha = 2$, the fusion performance is degraded. This indicates that the model prefers to extract object semantic features, resulting in the pixel-level

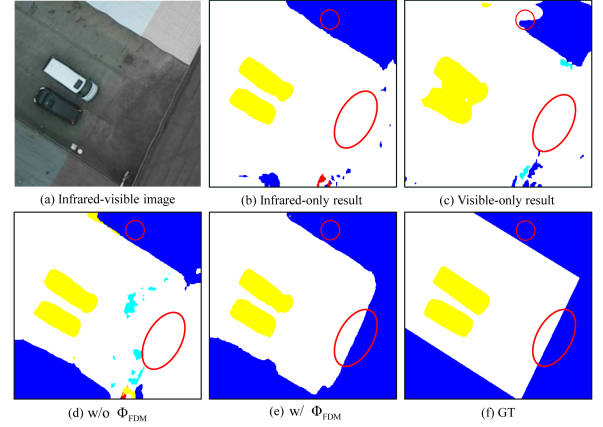


Fig. 10. Visual comparison of segmentation results produced by w/o Φ_{FDM} and w/ Φ_{FDM} on Potsdam dataset.

TABLE IV
EFFECTIVENESS OF DIFFERENT HYPERPARAMETERS α FOR FUSION
PERFORMANCE ON POTSDAM DATASET

	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$
$\alpha = 0.1$	6.872	35.009	17.179	6.464	1.687	1.720
$\alpha = 1$	6.944	37.616	16.946	6.370	1.800	1.752
$\alpha = 2$	6.776	32.631	16.469	6.175	1.554	1.683

The best result is in red and the second best one is in blue.

detail loss. Therefore, we take $\alpha = 1$ that ensures the model to achieve better fusion performance.

C. Comparison With State-of-The-Art Methods

We conduct qualitative and quantitative comparisons between our FreeFusion and the state-of-the-art methods, including YDTR [13], Tardal [1], U2Fusion [14], PIAFusion [15], SegMif [21], CDDFuse [16] and Text-IF [53].

1) Qualitative Comparison: Qualitative results of the different fusion methods on the Potsdam, WHU and MFNet datasets are shown in Figs. 11, 12, 13. Not only thermal information is presented in infrared image, whereas, the other characteristics, such as contrast, and texture information are also included. In the same way, visible images contain not only texture information, but also color, edge, and structural information. In this section, we compare the fusion results of different methods regarding the retention of vital features from source images.

In Fig. 11, visible images present color information with high saturation and infrared images present thermal information with high pixel intensity. As shown in the red boxes, YDTR, Tardal, U2Fusion, PIAFusion, SegMif and Text-IF cannot retain the color information of the visible images. We can clearly observe that the tree has low color saturation and blurred texture details. CDDFuse fuses the color information of the visible images, but loses high contrast thermal radiation information. Taking the object of soccer goal as an example, YDTR and Tardal cannot retain the thermal radiation information of the infrared image (e.g., door frame). In contrast, our method can fuse the vital information both in the infrared image such as high-intensity

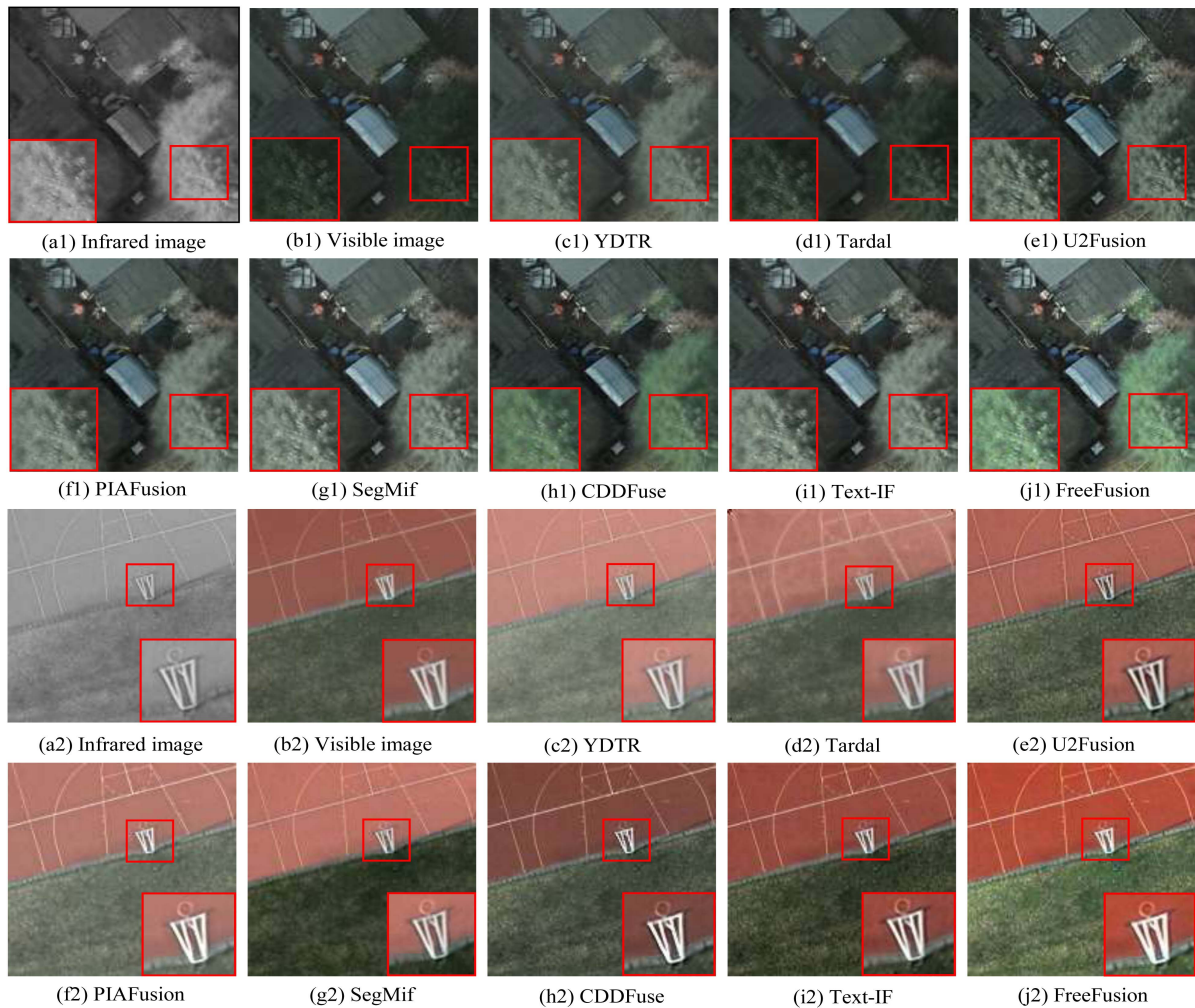


Fig. 11. Qualitative comparison of different fusion methods on Potsdam dataset.

thermal radiation information and in the visible image such as texture and color information.

In Fig. 12, infrared images present high contrast information and visible images present edge detail information. U2Fusion, CDDFuse, Text-IF and our method retain the edge details of the visible images. Taking the city (e.g., buildings) as an example, PIAFusion and SegMif can maximize fusion of edge details in the visible image. Tardal loses vital information of the visible image, which presents various degrees of the object ambiguity. In contrast, our fusion results can preserve structural and edge details from the source images.

In Fig. 13, the fusion results aim to retain objects and background information from the source images in the nighttime condition. As shown in red boxes, YDTR, U2Fusion and CDDFuse retain the structural features of the visible image, but lose the intensity information of the infrared image. Tardal, SegMif, PIAFusion and Text-IF cannot retain the structural information, resulting in the loss of object information (e.g., human's legs). Tardal cannot adaptively adjust the intensity distribution of the object region (e.g., pedestrian's pants) according to the texture information of the visible image, which produces fuzzy texture details. In contrast, our fusion result effectively integrates

TABLE V
QUANTITATIVE RESULTS OF DIFFERENT FUSION METHODS ON POTSDAM DATASET

Potsdam	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$
YDTR [13]	6.268	24.055	10.180	3.608	0.310	1.396
Tardal [1]	6.348	27.581	9.915	3.409	0.524	1.238
U2Fusion [14]	6.726	31.217	15.521	6.124	1.303	1.512
PIAFusion [15]	6.632	31.677	13.883	4.956	1.510	1.583
SegMif [21]	6.758	34.717	12.440	4.616	1.593	1.470
CDDFuse [16]	6.882	36.598	16.570	5.976	1.782	1.712
Text-IF [53]	6.777	33.814	14.533	5.435	1.601	1.624
Ours	6.944	37.616	16.946	6.370	1.800	1.752

The best result is in red and the second best one is in blue.

complementary information that clearly exhibits clearer texture details. Overall, our fusion result can adaptively fuse vital information, including significant characteristics and background texture information. Notice that, the image in Fig. 13 appears blurry due to its size of 320×320 .

2) *Quantitative Comparison:* Quantitative comparisons on Potsdam, WHU and MFNet datasets are shown in Tables V, VI, and VII. In Table V, our fusion results obtain

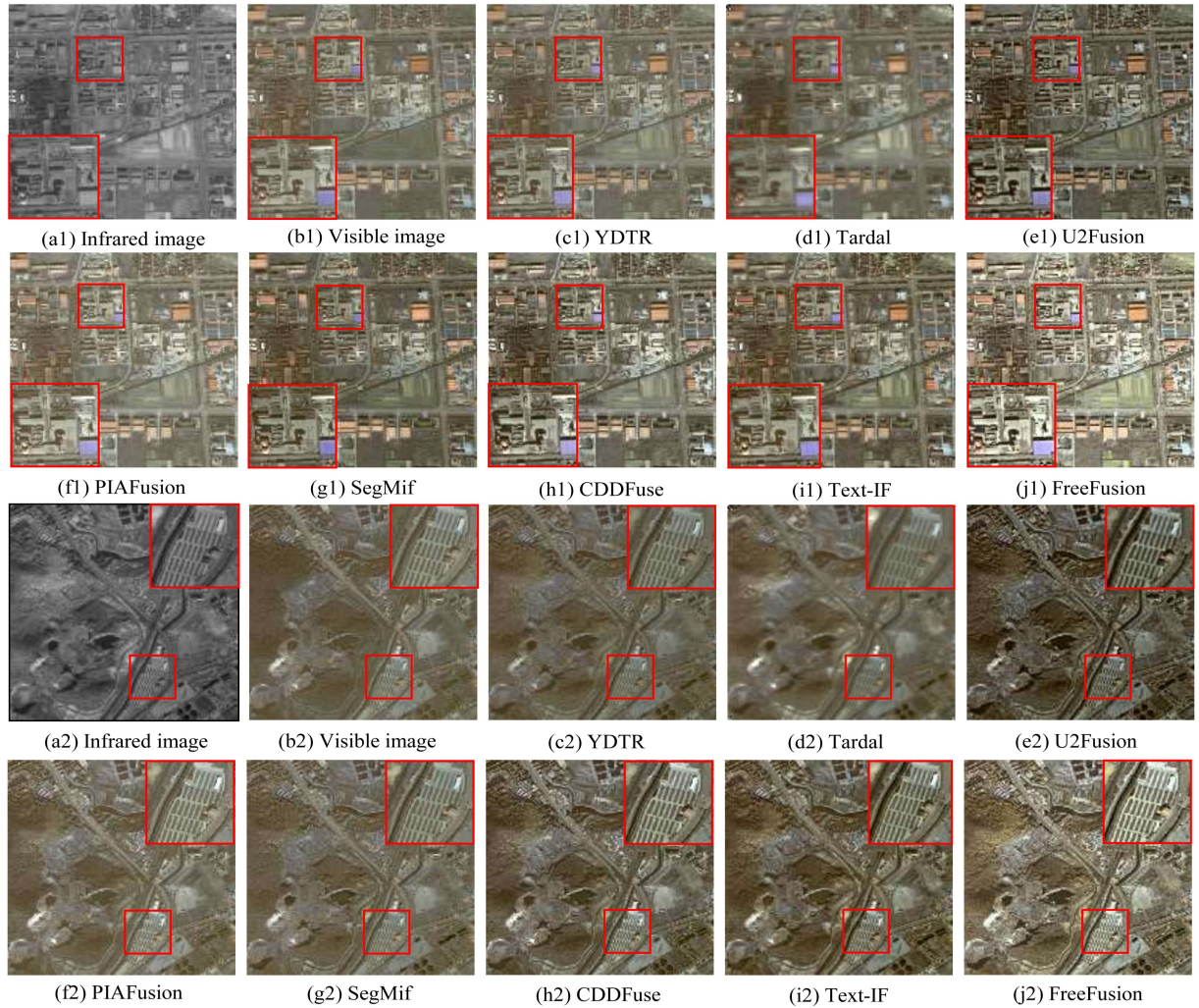


Fig. 12. Qualitative comparison of different fusion methods on WHU dataset.

TABLE VI

QUANTITATIVE RESULTS OF DIFFERENT FUSION METHODS ON WHU DATASET

WHU	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$
YDTR [13]	5.878	16.617	14.365	5.01	0.829	1.015
Tardal [1]	6.781	32.812	15.465	5.867	1.033	0.796
U2Fusion [14]	6.766	29.826	24.865	10.041	1.691	1.004
PIAFusion [15]	6.123	19.938	18.303	6.428	1.187	1.043
SegMif [21]	6.304	22.582	20.262	7.380	1.257	1.049
CDDFuse [16]	6.737	29.773	26.558	9.231	1.782	1.202
Text-IF [53]	6.653	27.614	23.578	8.747	1.584	1.163
Ours	6.954	34.576	33.147	11.744	1.896	1.195

The best result is in red and the second best one is in blue.

the best metric scores, improving EN/SD/SF/AG/SCD/VIF by 0.9% /2.8% /2.3% /4.0% /4.0% /1.0% /2.3% relative percentage than the sub-optimal results. In Table VI, our fusion results generally outperform the other methods, improving 2.6% /5.4% /24.8% /17% /6.4% relative percentage than the sub-optimal method in EN/SD/SF/AG/SCD. In Table VII, our fusion results achieve competitive performance, improving 2.6% /2.2%

TABLE VII

QUANTITATIVE RESULTS OF DIFFERENT FUSION METHODS ON MFNET DATASET

MFNet	EN $\uparrow$	SD $\uparrow$	SF $\uparrow$	AG $\uparrow$	SCD $\uparrow$	VIF $\uparrow$
YDTR [13]	5.711	22.047	7.457	2.370	1.258	0.657
Tardal [1]	6.632	36.838	11.761	3.527	1.544	0.783
U2Fusion [14]	6.237	33.610	12.082	4.207	1.581	0.777
PIAFusion [15]	6.398	36.149	12.562	4.239	1.481	0.940
SegMif [21]	6.507	37.871	11.757	4.056	1.578	0.830
CDDFuse [16]	6.348	36.900	12.087	3.850	1.541	0.944
Text-IF [53]	6.623	39.467	12.581	4.220	1.591	1.055
Ours	6.702	39.008	12.853	4.545	1.800	0.937

The model inference time is measured on an NVIDIA GeForce RTX 4090. The best result is in red and the second best one is in blue.

/7.2% /13.9% than the second best values in EN/SF/AG/SCD. Quantitative results indicate that our fusion results retain richer texture details, higher contrast and larger gradients from the source images. As shown in Table VIII, our method achieves the second-ranked inference time that consumes 0.0049 s. Besides, our method incurs a total of 99.08 G FLOPs, where $\Phi_{E,j}^{ir}$,

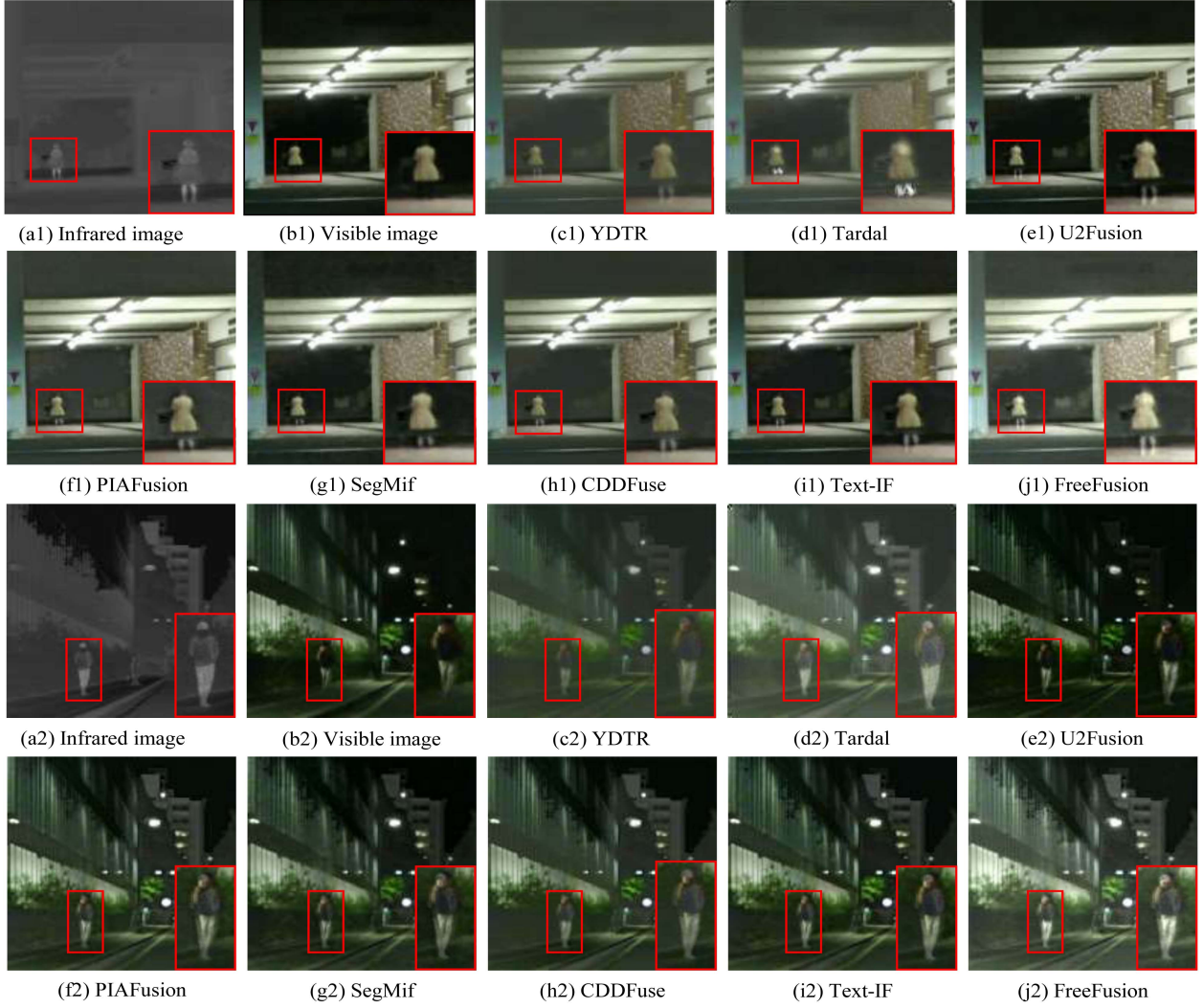


Fig. 13. Qualitative comparison of different fusion methods on MFNet dataset.

 TABLE VIII
 EFFICIENCY ANALYSIS OF INFERENCE TIME AND FLOPS ON POTSDAM DATASET

	YDTR [13]	Tardal [1]	U2Fusion [14]	PIAFusion [15]	SegMif [21]	CDDFuse [16]	Text-IF [53]	Ours
Time(S)↓	0.0175	0.0063	0.0075	0.0016	0.0124	0.0575	0.0787	0.0049
FLOPs(G)↓	32.16	30.38	135.06	120.33	117.90	61.20	507.59	99.08

The model inference time is measured on an NVIDIA GeForce RTX 4090. The best result is in red and the second best one is in blue.

$\Phi_{E,j}^{vis}$, Φ_D and DIF_j account for 21.01G, 21.01G, 10.30G and 46.76G, respectively.

D. Generalization Experiment

To validate the generalization ability of our FreeFusion, we select the fusion model trained on the WHU dataset, and test it directly on the M³FD and LLVIP datasets.

The fusion results in M³FD and LLVIP datasets are shown in Fig. 14. Intuitively, the infrared image presents thermal radiation information with high pixel intensity (the first group), and the visible image presents abundant texture detail information (the second group). As shown in the red boxes (c1,

d1, f1 and h1), YDTR, Tardal, PIAFusion and CDDFuse fail to preserve the intensity distribution of salient object region (e.g., white cloud). Additionally, Tardal overemphasizes the preservation of thermal target information for infrared images, which causes the fusion results not to adaptively adjust the thermal radiation information and texture information. As shown in the red boxes (c2-e2 and h2), YDTR, Tardal, U2Fusion and CDDFuse lose some texture details of the visible image, accompanied by ambiguity (e.g., bollard and pavement). In contrast, our method freely preserves the texture details of visible images and the object intensity information of infrared images. The advantage stems from that our method is not using hand-crafted fusion losses, which makes the fusion image

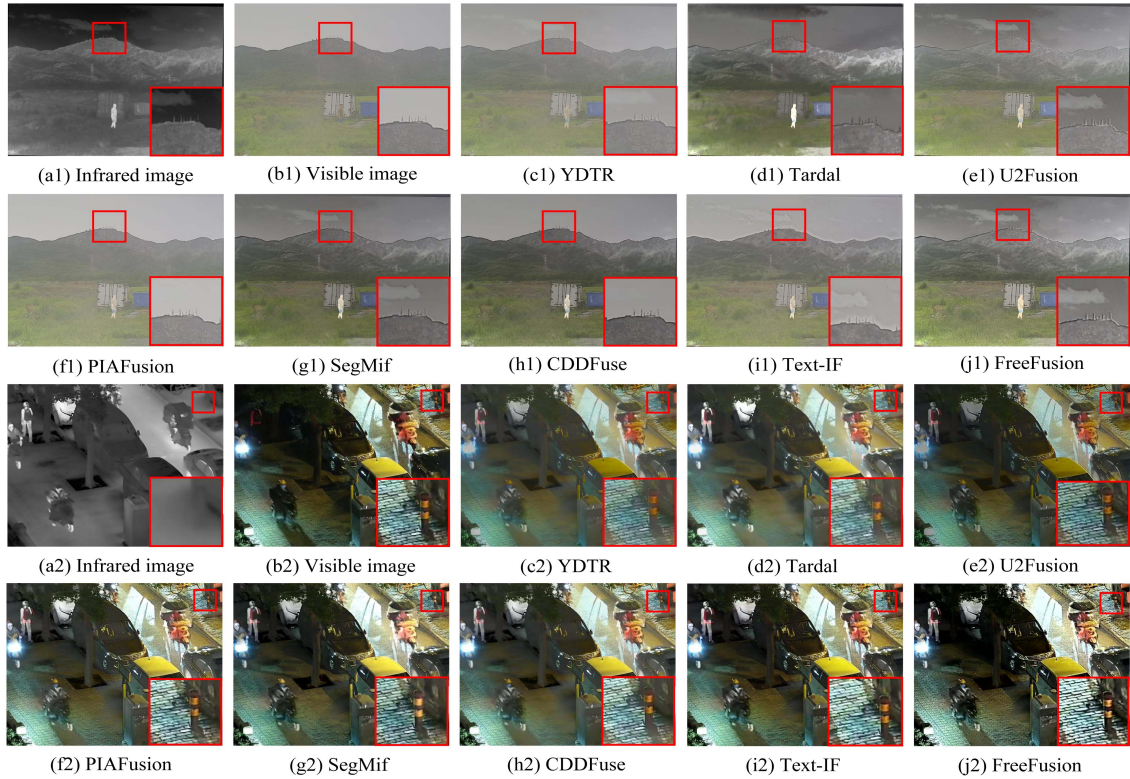


Fig. 14. Qualitative comparison of different fusion methods on LLVIP (the first group) and M³FD (the second group) datasets.

TABLE IX
QUANTITATIVE RESULTS OF DIFFERENT FUSION METHODS ON M³FD DATASET

M ³ FD	EN↑	SD↑	SF↑	AG↑	SCD↑	VIF↑
YDTR [13]	6.574	28.318	10.752	3.454	1.549	0.624
Tardal [1]	7.168	43.231	12.857	4.237	1.581	0.594
U2Fusion [14]	6.686	29.484	11.342	4.120	1.620	0.621
PIAFusion [15]	6.910	37.512	16.061	5.304	1.531	0.843
SegMif [21]	7.065	39.658	14.847	4.950	1.827	0.770
CDDFuse [16]	6.939	37.917	15.497	5.016	1.667	0.774
Text-IF [53]	6.990	38.009	16.473	5.456	1.550	0.880
Ours	7.228	45.382	20.171	7.128	1.916	0.769

The best result is in red and the second best one is in blue.

TABLE X
QUANTITATIVE RESULTS OF DIFFERENT FUSION METHODS ON LLVIP DATASET

LLVIP	EN↑	SD↑	SF↑	AG↑	SCD↑	VIF↑
YDTR [13]	6.740	35.083	9.915	2.487	1.126	0.581
Tardal [1]	7.373	52.557	14.003	3.730	1.361	0.640
U2Fusion [14]	6.708	36.287	11.900	3.675	1.282	0.651
PIAFusion [15]	7.345	49.733	16.865	5.039	1.535	0.900
SegMif [21]	7.323	51.491	15.719	4.476	1.688	0.907
CDDFuse [16]	7.334	49.404	15.248	4.100	1.583	0.880
Text-IF [53]	7.397	52.422	18.358	5.423	1.597	0.990
Ours	6.969	56.786	20.81	5.975	1.749	0.776

The best result is in red and the second best one is in blue.

more comprehensively preserve all the vital features of source images.

Qualitative comparisons are shown in Tables IX and X. In Table IX, compared with the second ranked method, our fusion results improve EN, SD, SF, AG, and SCD by 0.83%, 4.74%, 18.33%, 23.46%, and 4.65%, respectively. In Table X, our fusion results achieve the best performance on SD, SF, AG, and SCD, improving 7.45%, 11.78%, 9.24%, and 3.49% over the second best results, respectively. Overall, our FreeFusion achieves superior performance on most evaluation metrics.

E. Performance on High-Level Task

1) *Evaluation of Infrared-Visible Semantic Segmentation:* A better fusion result can effectively assist the segmentation task to improve the segmentation accuracy. We further verify the effectiveness of our method by evaluating the segmentation accuracy with the infrared and visible fusion images on the Potsdam, WHU, and MFNet datasets. The fusion results generated by our method and the other methods are retrained to semantic segmentation network with the same hyperparameter configuration. The semantic segmentation baseline SegFormer [54] is adopted.

Qualitative results of different methods are shown in Fig. 15. In Fig. 15 (the first group), our FreeFusion is able to effectively highlight information about vegetation, roadways, and car compared to the other methods. For example, YDTR, Tardal, U2Fusion, PIAFusion and Text-IF fail to preserve the color and texture features from the visible image, so as that segmentation network is unable to recognize "Low vegetation" and "Impervious surface" areas in the scene. In Fig. 15 (the second group), our FreeFusion learns complementary characteristics of

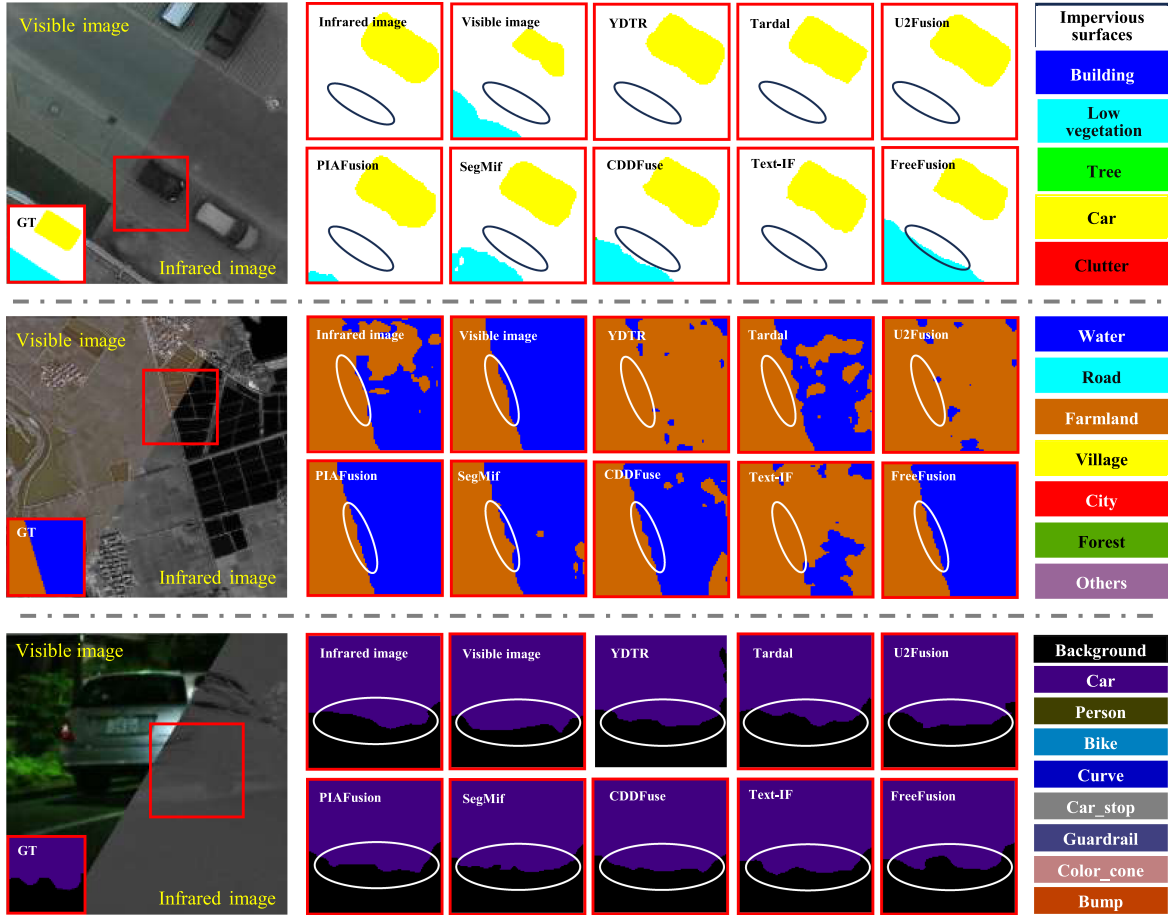


Fig. 15. Qualitative comparison of semantic segmentation based on different fusion methods on Potsdam (the first group), WHU (the second group) and MFNet datasets (the third group).

the infrared and visible images to improve the segmentation accuracy. For example, the visible images highlight the background information that recognizes the categories "water" and "farmland". The existing methods generate fusion results that lose the vital features (e.g., edge details) of the visible images, leading to inaccurate categorization. In Fig. 15 (the third group), our FreeFusion is able to recognize the complete structure of "Car" in the extreme darkness. Since SegMif and CDDFuse lose the thermal radiation information of the infrared image, the fusion results cannot help segment the car contours (e.g., wheels). In contrast, our method generates fusion results that retain vital features from the source images (e.g., gradients, objects, contours, *etc.*), thereby improving the segmentation accuracy.

Table XI reports the quantitative results. Compared to the second best values on Potsdam and WHU datasets, our FreeFusion improves 0.92% / 0.48% relative percentage on mDice and 0.46% / 0.32% relative percentage on mAcc metric. Our FreeFusion ranks the first of mDice on MFNet dataset, improving 0.29% relative percentage compared to the second best value. This proves that our FreeFusion generates fusion images that effectively enhance both object and background information.

2) *Evaluation of Infrared-Visible Object Detection:* We study the impact of the fusion results on the object detection

TABLE XI
QUANTITATIVE COMPARISONS OF SEMANTIC SEGMENTATION BASED ON DIFFERENT FUSION METHODS ON POTSDAM, WHU AND MFNET DATASETS

	Potsdam		WHU		MFNet	
	mDice↑	mAcc↑	mDice↑	mAcc↑	mDice↑	mAcc↑
Visible	73.99	73.07	53.25	51.79	46.03	41.84
Infrared	69.59	68.24	53.23	51.54	41.76	40.37
YDTR [13]	74.23	73.36	52.61	50.64	46.04	39.78
Tardal [1]	72.53	71.03	52.95	51.92	45.51	40.24
U2Fusion [14]	73.68	72.19	53.78	51.74	47.47	43.17
PIAFusion [15]	75.15	73.76	51.74	50.50	45.57	40.05
SegMif [21]	74.62	73.27	52.84	52.18	45.1	39.72
CDDFuse [16]	74.49	73.33	53.43	51.57	44.83	40.32
Text-IF [53]	74.61	73.68	53.65	52.61	46.82	41.47
Ours	75.31	74.02	54.04	52.78	47.61	42.13

The best result is in red and the second best one is in blue.

performance on M³FD and LLVIP datasets. Specifically, we use the fusion results generated by our method and SOTA methods to retrain the detection model of YOLOv5.¹

Fig. 16 shows the qualitative detection results of different methods. In the first group, existing fusion methods fail to adequately preserve the complete structure (e.g., bus) of the visible images and focus overly on thermal radiation information, resulting in inaccurate detection of distant objects. In the second

¹ <https://github.com/bupt-ai-cz/LLVIP/tree/main/yolov5>

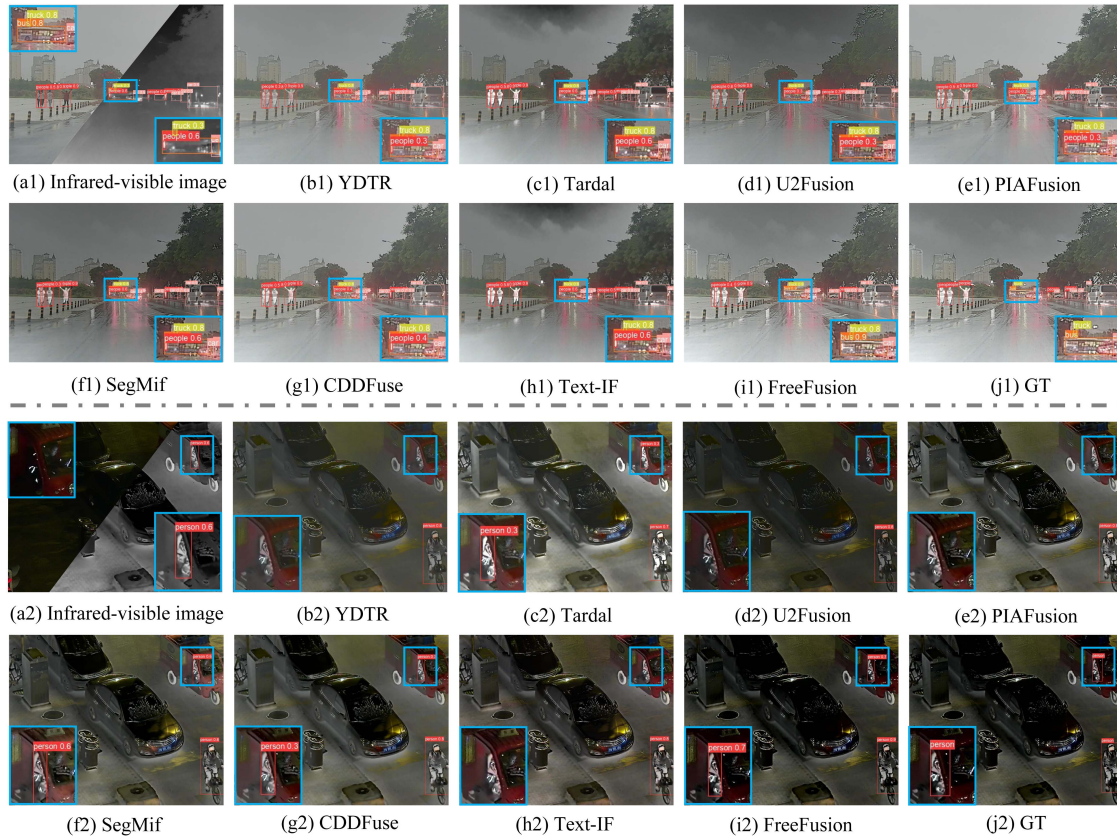


Fig. 16. Qualitative comparison of object detection based on different fusion methods on M³FD (the first group) and LLVIP (the second group) datasets.

TABLE XII
QUANTITATIVE COMPARISONS OF OBJECT DETECTION BASED ON DIFFERENT
FUSION METHODS ON M³FD AND LLVIP DATASETS

	M ³ FD	LLVIP
	mAP _{50→95} (%)↑	mAP _{50→95} (%)↑
Visible	56.6	52.4
Infrared	50.6	66.0
YDTR [13]	58.2	64.3
Tardal [1]	57.2	63.2
U2Fusion [14]	58.7	64.9
PIAFusion [15]	58.2	64.4
SegMif [21]	58.6	64.7
CDDFuse [16]	58.5	65.1
Text-IF [53]	58.9	64.9
Ours	62.6	66.4

The best result is in red whereas the second best one is in blue

group, YDTR, U2Fusion, PIAFusion and Text-IF fail to adequately integrate texture details and high contrast information, resulting in difficulty in detecting “person”. Although the fusion results generated by other methods can detect object “person”, the model’s confidence scores for detecting the object “person” are still lower than our FreeFusion. In contrast, our FreeFusion generates detection-friendly fusion results that help the object detection model achieve superior detection performance in different scenarios.

Table XII reports the quantitative results. We can see that our FreeFusion obtains the best detection results, which outperform

the second-ranked method by 5.91% and 1.96% in terms of mAP_{50→95}(%), respectively.

V. CONCLUSION

This paper presents a cross reconstruction learning method for infrared and visible image fusion without using any hand-designed fusion loss. Especially, the cross-reconstruction learning is designed to decouple the fusion features to generate another modality features, thereby crossly reconstructing the source images. This ensures that the network adaptively fuses important features and achieves the domain alignment of two modalities during the decoupling and reconstruction process. Moreover, the dynamic interaction fusion is built that addresses the dilemma of the semantic mismatch between two different-level tasks. Thus, the feature correlation matrix of two tasks is designed to achieve the mapping alignment, and then enables their mutual guidance in the fusion process. Extensive qualitative and quantitative experiments on three public datasets demonstrate that our FreeFusion achieves the superior fusion performance compared to existing state-of-the-art methods, and facilitates the downstream segmentation accuracy.

FreeFusion achieves the two modality domain alignment through the cross-reconstruction learning. However, since the two modality features are not explicitly aligned, mapping source images with large domain discrepancy to a same domain space is challenging. In future work, we attempt to utilize textual prompts

to provide an explicit description from source images, thereby guiding the two modalities to align in the same text space for eliminating domain discrepancy.

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